



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2018

NAME : _____

CT CLASS : _____

H1 BIOLOGY

8876/02

Paper 2

2 hours

READ THESE INSTRUCTIONS FIRST

Write your Name and CT class on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Write your answers on separate answer paper provided.

At the end of the examinations,

1. hand in section A and the one question you attempted from section B separately;
2. fasten all your work securely;
3. enter the number of the section B question you have answered in the grid opposite.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Section B	
Total	

Section A [45 marks]

Answer all questions.

- 1 In eukaryotic cells, the degradation of mRNA is an essential part of the regulation of gene expression. It can be controlled in response to developmental, environmental, and metabolic signals. mRNA hydrolysis is catalysed by numerous types of nucleases, such as the endonuclease Ribonuclease A (RNAse A), shown in Fig. 1.1.

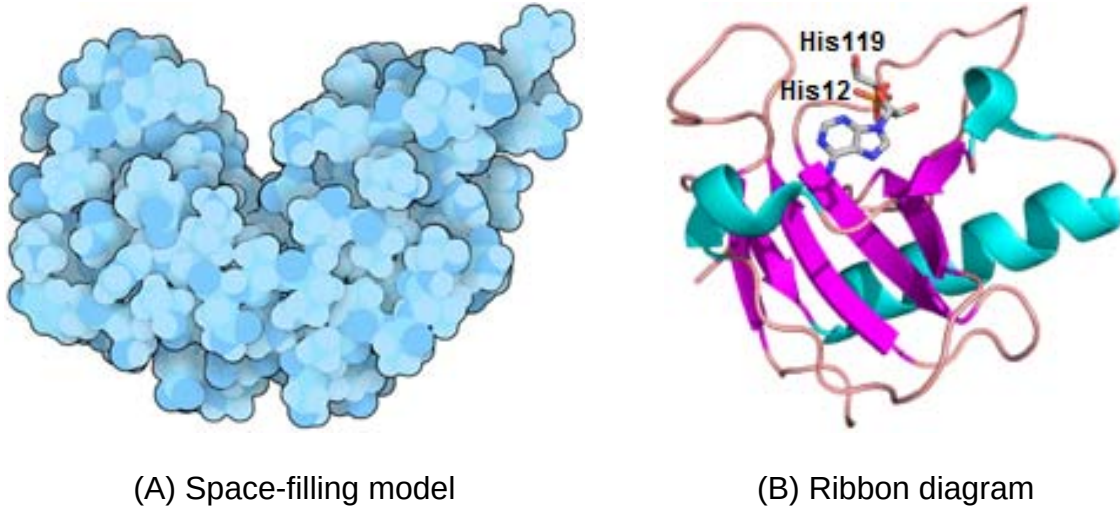


Fig. 1.1

- (a) Using a labelled and annotated diagram, illustrate the hydrolysis of the bond catalysed by RNAase.
(A monomer has been drawn for you.)

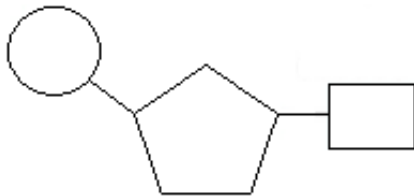


Fig. 1.1B shows two important catalytic residues within the active site of RNase A, which are His12 and His119.

- (b) Explain how these two histidines, which are in position 12 and 119 of the 124 amino acid sequence, are brought together in the active site of the enzyme.

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Fig. 1.2 shows the structure of histidine and phenylalanine.

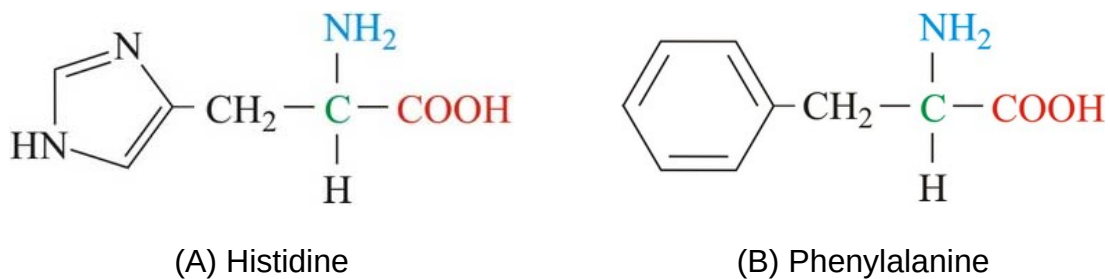


Fig. 1.2

- (c) Predict and explain how the catalytic activity of RNase would be affected if both histidines were replaced by phenylalanines.

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Ribonuclease A is also the enzyme that digests RNA in our food. It is released by the pancreas of most mammals and some reptiles. Because it is small, stable, and easily purified, ribonuclease has been an important enzyme in biochemical research. In one procedure for purifying RNase A from bovine pancreas, extracts are treated with mild sulfuric acid and then heated almost to boiling, leaving RNase A as the only protein remaining intact.

(d) Suggest why RNase A has such a stable structure and why it needs to be very stable. [2]

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[Total: 9]

2 Table 2.1 shows some mRNA codons and the amino acids for which they code.

amino acid	abbreviation	mRNA					
glutamic acid	glu	GAA	GAG	–	–	–	–
phenylalanine	phe	UUU	UUC	–	–	–	–
lysine	lys	AAA	AAG	–	–	–	–
proline	pro	CCA	CCC	CCG	CCU	–	–
threonine	thr	ACA	ACC	ACG	ACU	–	–
valine	val	GUA	GUC	GUG	GUU	–	–
cysteine	cys	UGC	UGU	–	–	–	–
arginine	arg	CGC	CGA	CGU	CGG	AGA	AGG

Table 2.1

Fig. 2.2 shows the sequence of three amino acids in the human lysozyme polypeptide.

amino acids	arg	cys	glu
tRNA			
Template strand DNA			

Fig. 2.2

(a) Use the information in Table 2.1 to fill in any one possible tRNA anticodon and corresponding DNA sequence for each amino acid shown in Fig. 3.2. Write your answers in Fig. 2.2. [1]

(b) Explain what an “anticodon” is and its significance in the synthesis of proteins.

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A student drew how DNA is replicated as shown in Fig. 2.3.

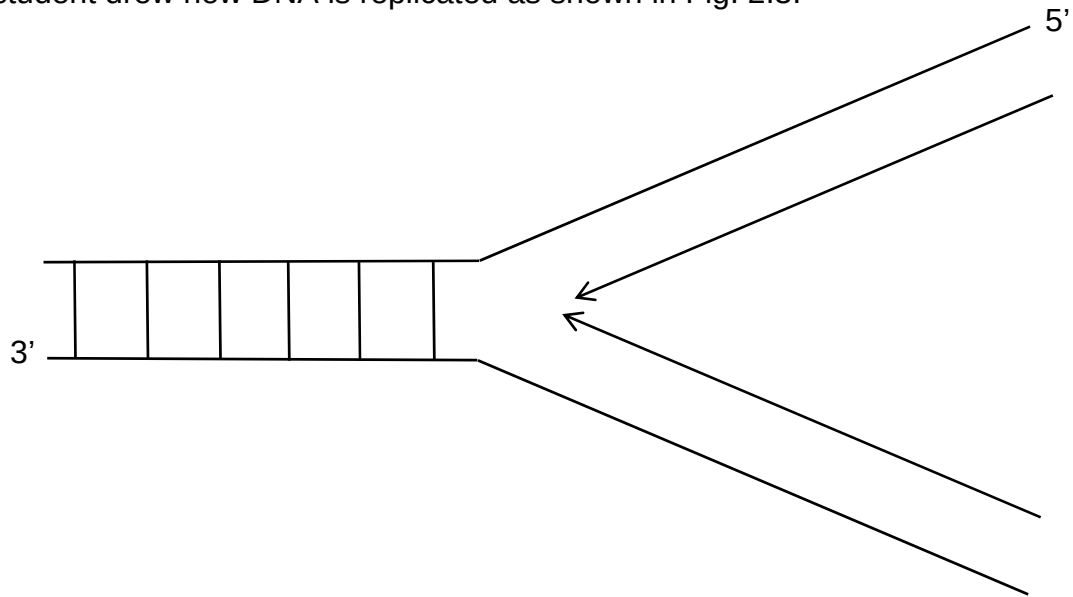


Fig. 2.3

(c) Evaluate the accuracy of the student's drawing.

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[Total: 7]

3 (a) Structure **Q** in Fig. 3.1 is a cell structure which is involved in nuclear division.

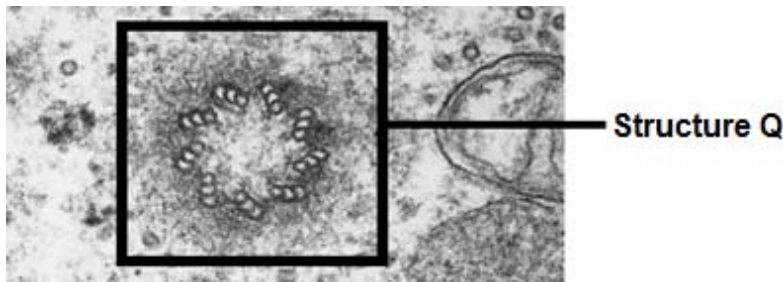


Fig. 3.1

(i) Identify structure **Q** and outline its role in nuclear division.

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 [2]

(ii) Compare the structure of **Q** and a ribosome.

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 [3]

(b) Chickpeas contain a lipase inhibitor that prevents the activity of the enzyme in the digestion of fats. There are two forms of lipase inhibitors – inhibitor **W** and inhibitor **X**.

Homozygous plants are known to produce one type of lipase inhibitor, depending on the allele which they are homozygous for. A heterozygote plant with both types of alleles, on the other hand, will have two types of lipase inhibitor, inhibitor **W** and inhibitor **X**. A third recessive allele does not code for any lipase inhibitor.

(i) State and explain the mode of inheritance for the lipase inhibitor in the chickpeas.

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 [2]

A second character, seed texture, is controlled by another gene located on a different chromosome and is controlled by two alleles. Smooth seed-coat, **T**, is dominant over wrinkled seed-coat, **t**.

Two chickpea plants were crossed. One of them contains only inhibitor **X** and has smooth seed-coats. Their seeds were collected and counted, as summarised in Table 3.2.

Inhibitor(s) present in seed	Number of seeds	Seeds with smooth seed-coat / %
W and X	12	50
W	14	50
X	22	50

Table 3.2

- (ii) Using suitable symbols, draw a genetic diagram to explain the results of this cross, as shown in Table 3.1.

[5]

[Total: 12]

- 4 In an experiment to investigate the effect of various environmental factors on seed germination, germinating soybeans were exposed to different temperatures and light intensities. Fig. 4.1 shows the experimental set-up.

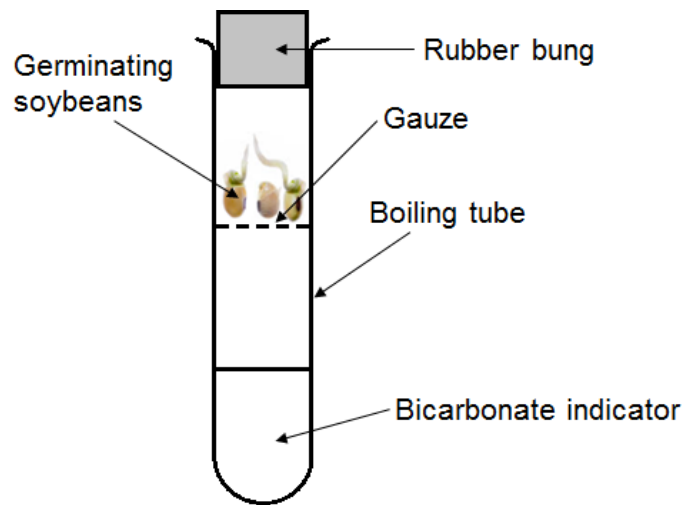


Fig. 4.1

Bicarbonate indicator is a pH indicator. It changes colours as the pH changes. At the start of the experiment, the bicarbonate indicator is at pH 7.

The set-up is left under different temperatures and light intensities for one hour, and the change in pH is recorded. Table 3.1 shows the results.

Temperature / °C	Light intensity	pH of bicarbonate indicator after 1 hour
10	High	6.5
10	Low	6.5
20	High	6.0
20	Low	6.0
30	High	5.0
30	Low	5.0
40	High	4.5
40	Low	4.5

Table 4.1

- (a) (i) Explain the reason for the drop in pH of the bicarbonate indicator.

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- (ii)** With reference to the information provided, explain the effect of light intensity on the germination of soybeans.

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Studies on cancer cells found that fast-growing cancer cells require much more energy than normal cells, which explains the much higher rate of glucose uptake into cancer cells. However, it is also found that, unlike normal cells, the higher glucose uptake reduces oxygen uptake into cancer cells. This respiratory inhibition is known the Crabtree effect.

- (b) (i)** Besides the need for more energy for cell division, explain why cancer cells utilise glucose at a much higher rate than normal cells to produce energy.

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- (ii)** Cancer cells undergo anaerobic respiration in the same way mammalian muscle cells do. Compare the difference between anaerobic respiration in cancer cells and yeast cells.

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- (c)** Explain why ATP is regarded as the universal energy currency.

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[Total: 9]

- 5** It is predicted that by year 2100, global warming may be a lot more severe than expected should greenhouse emissions continue to proceed at the current rate. Global temperatures can potentially increase by as much as 5°C.

Insects are known to be sensitive to temperature changes and they face risks associated with global warming. A study conducted on 50 fruit flies suggested that limiting global warming to 1.5°C would save the vast majority of the world's insect species from climate change.

(a) Outline how global warming can impact insects negatively.

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Reducing the risk to insects is particularly important, the research team say, as the effect of climate change on them can potentially implicate human beings.

(b) Suggest how the impact of climate change on insects can potentially implicate humans.

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(c) Some scientists, however, believe that insects could evolve to become more heat-tolerant in the future. Explain how natural selection may lead to the evolution of more heat-tolerant insect species.

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[Total: 8]

Section C [15 marks]

Answer **one** question in this section. Each question has a part (a) and part (b).

Write your answers on the writing paper provided, with each part on a fresh sheet of paper.

Your answers should be in continuous prose, where appropriate.

Your answers should be accompanied by large, clearly labelled diagrams, where appropriate.

- 6 (a)** The Taq polymerase is a thermostable DNA polymerase named after the thermophilic bacterium *Thermus aquaticus* from which it was originally isolated from.

With reference to different levels of protein structure, explain how it is possible for both human DNA polymerase and the Taq polymerase to act on the same substrates, but for the Taq polymerase to have a higher optimum temperature

[6]

- (b)** Explain, using examples, the significance of the fluid mosaic model to the functions of membranes at the cell surface or within cells.

[9]

[Total: 15]

- 7 (a)** Explain how genes are inherited from one generation to the next via the germ cells or gametes.

[6]

- (b)** Bone marrow contains many stem cells. Some of these stem cells are responsible for the replacement of red blood cells. During the production of red blood cells, a series of changes occur to the cell structure.

Describe the features of the stem cells responsible for production of red blood cells and outline the main events that must occur within the cell to result in the formation of the red blood cell.

[9]

[Total: 15]